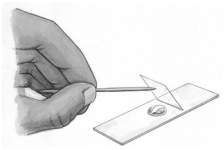
Worksheet 3.3

Preparing a stained wet mount

Student: ..... Class: .....

Jumbled steps

1. The following diagrams show the steps used to prepare a wet mount of onion epidermis, but the order has been jumbled. Place the code letters in the correct sequence.



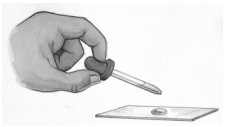
A. Lower a coverslip over the onion skin and water droplet.



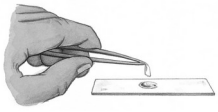
B. Peel a section of onion epidermis.



C. Soak up excess water with a tissue.



D. Add a drop of water to a microscope slide.



E. Place onion skin in drop of water on the slide.

Correct sequence: .....

Staining cells

2. It is much easier to see the internal structure of cells by staining the tissue. The following materials are available to prepare a stained wet mount of onion epidermis.

dropping bottle of methylene blue or other similar stain, microscope, microscope slide, coverslip, small section of a peeled onion, forceps, toothpick, dropping bottle of water

(a) Write a method for your investigation.

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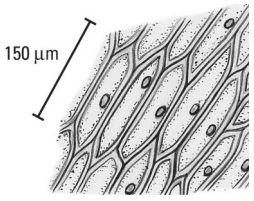
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(b) The drawing below left was made of the stained onion epidermal tissue.



(i) The nucleus and cell walls take up the methylene blue stain. Use a blue pencil to colour these components in the drawing.

(ii) Use the scale to estimate the greatest width of a cell.

(iii) The experiment was repeated with iodine stain, which strongly stains starch grains a deep blue-black colour. Use coloured pencils to colour the starch grains, which are scattered in the cytoplasm in the cells at the right.



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